

M05 PROTECTED PRACTICE REVIEW

Open only after a genuine saved PBIX attempt

Diagnose -> close review -> changed model task -> append evidence.

PBM0 review - Install, orient and import

1. Strong attempt should show

PBIX saved; all intended tables imported through Transform Data; learner can name the four working areas correctly.

2. Expected reasoning

Core teacher model: Source workbook -> Navigator -> Transform Data -> Power Query Editor -> Close & Apply -> Model view -> temporary validation visual. Power BI Desktop has four working areas you must distinguish: Report view for visuals, Data view for loaded data inspection, Model view for tables/relationships, and Power Query Editor for repeatable preparation before load. A PBIX is the model/report project file. Use Get Data -> Excel -> Transform Data so you inspect the source before loading it blindly.

3. What to compare

Compare source/prepared row counts, grain, key assumptions, relationship/filter path and control totals before comparing screenshots or clicks. Equivalent valid Power BI routes are acceptable when the validation contract is satisfied.

4. Likely repair

If the wrong sheet/query was loaded, remove the unintended query and reconnect through Transform Data. If Power BI auto-created relationships, do not trust them yet; PBM4 will rebuild relationships deliberately.

5. Fresh retry

Close this review. Use the changed transfer task. Save a new PBIX/version/evidence row. Only a fresh successful retry can upgrade mastery.

PBM1 review - Power Query interface and profiling

1. Strong attempt should show

Profiling note exists before transformations and distinguishes duplicate-key suspicion from proven duplicate records.

2. Expected reasoning

Core teacher model: For each query: Grain -> Rows -> Required blanks -> Key uniqueness -> Types -> Business validity -> Consistency -> only then transformations. Profile before transforming. For every table, state its intended grain, row count, required-key completeness, one-side key uniqueness, type problems, suspicious business values and formatting inconsistencies. Applied Steps are an ordered preparation pipeline; each change should be explainable and reversible by editing the step rather than manually changing the raw workbook.

3. What to compare

Compare source/prepared row counts, grain, key assumptions, relationship/filter path and control totals before comparing screenshots or clicks. Equivalent valid Power BI routes are acceptable when the validation contract is satisfied.

4. Likely repair

If profiling feels vague, start with grain + row count + required keys. For a suspected duplicate, compare the full business row and normalized key before deleting anything.

5. Fresh retry

Close this review. Use the changed transfer task. Save a new PBIX/version/evidence row. Only a fresh successful retry can upgrade mastery.

PBM2 review - Transform safely and set data types

1. Strong attempt should show

Every row-count change is explained; invalid/unmatched values remain visible; no raw-sheet manual edit is required.

2. Expected reasoning

Core teacher model: Raw value -> justified transformation -> explicit data type -> quality check -> prepared value. Every row-count change gets a reason. A safe transformation is repeatable, justified and testable. Trim formatting noise, standardize only confirmed equivalent values, set business-appropriate types, remove an exact duplicate only after proving it is exact at the stated grain, and keep unresolved unmatched/invalid values visible. Type validity and business validity are different: 1.20 is numeric but may violate a 0%-100% discount rule.

3. What to compare

Compare source/prepared row counts, grain, key assumptions, relationship/filter path and control totals before comparing screenshots or clicks. Equivalent valid Power BI routes are acceptable when the validation contract is satisfied.

4. Likely repair

If a type change creates errors, inspect the original values and isolate the failure instead of replacing all errors with zero/blank. If row count changed unexpectedly, step backward through Applied Steps until the count changes.

5. Fresh retry

Close this review. Use the changed transfer task. Save a new PBIX/version/evidence row. Only a fresh successful retry can upgrade mastery.

PBM3 review - Fact tables, dimensions, grain and star schema

1. Strong attempt should show

Every table has a written grain; intended one-side keys are unique; fact is central in a recognizable star.

2. Expected reasoning

Core teacher model: Fact (many event rows) <- foreign keys -> Dimensions (unique descriptive keys). Grain is written before relationship creation. A fact table records events/measurements at a consistent grain and usually has many rows. Dimensions describe entities used to group/filter and require a unique key on the one side. A star schema keeps the fact in the center with Product/Customer/Region/Date-like dimensions around it. Model clarity begins with one sentence describing the grain of every table.

3. What to compare

Compare source/prepared row counts, grain, key assumptions, relationship/filter path and control totals before comparing screenshots or clicks. Equivalent valid Power BI routes are acceptable when the validation contract is satisfied.

4. Likely repair

If the dimension key is not unique, do not create many-to-many as a shortcut. Identify whether whitespace, duplicate business entities or wrong grain caused the duplicate.

5. Fresh retry

Close this review. Use the changed transfer task. Save a new PBIX/version/evidence row. Only a fresh successful retry can upgrade mastery.

PBM4 review - Relationships, cardinality and filter direction

1. Strong attempt should show

Relationship list shows 1:* and single direction; a temporary dimension filter changes only matching fact rows.

2. Expected reasoning

Core teacher model: Dimension[Key] (1) -> Fact[ForeignKey] (*) with single-direction filter flow. Duplicate dimension key = stop and investigate; unmatched fact key = quality issue to quantify. Relationships encode business logic. A one-to-many relationship requires uniqueness on the intended one side; unmatched foreign keys on the fact side are a different issue. Default to single-direction filtering from dimension to fact. If Power BI refuses 1:*, investigate duplicate/blank/type-mismatched one-side keys instead of switching to many-to-many to make the warning disappear.

3. What to compare

Compare source/prepared row counts, grain, key assumptions, relationship/filter path and control totals before comparing screenshots or clicks. Equivalent valid Power BI routes are acceptable when the validation contract is satisfied.

4. Likely repair

If a relationship cannot be 1:*, inspect one-side duplicates/blanks/types. If a filter does not propagate, verify active relationship, direction and matching key values before enabling bidirectional filtering.

5. Fresh retry

Close this review. Use the changed transfer task. Save a new PBIX/version/evidence row. Only a fresh successful retry can upgrade mastery.

PBM5 review - Date dimension and date relationships

1. Strong attempt should show

Date is unique/nonblank/continuous for the required range; Date -> fact relationship is active and correctly directed.

2. Expected reasoning

Core teacher model: Date[Date] (1) -> Fact[Business Date] (*). Date dimension has continuous dates and separate Month + Month Number attributes. A Date dimension gives the model one consistent calendar. It should contain one row per date, a unique/nonblank Date key, no gaps in the required range and useful attributes such as Year, Month, Month Number, Quarter and Year-Month. Connect Date[Date] 1:* to the fact date and keep Month Number so month names can sort chronologically.

3. What to compare

Compare source/prepared row counts, grain, key assumptions, relationship/filter path and control totals before comparing screenshots or clicks. Equivalent valid Power BI routes are acceptable when the validation contract is satisfied.

4. Likely repair

If Date does not cover the fact dates or contains duplicates/gaps, stop and repair/replace the calendar range before building time analysis. Do not create DAX time intelligence in this module.

5. Fresh retry

Close this review. Use the changed transfer task. Save a new PBIX/version/evidence row. Only a fresh successful retry can upgrade mastery.

PBM6 review - Model validation and trust checks

1. Strong attempt should show

Validation log contains independent checks and reconciles the protected gross/control total without hiding quality issues.

2. Expected reasoning

Core teacher model: Trust = row reconciliation + key checks + unmatched-key checks + type/business checks + control total + filter-path proof. A model is trustworthy only when multiple independent checks agree. Reconcile source/prepared row counts, fact-grain uniqueness, dimension-key uniqueness, unmatched foreign keys, conversion/business-rule issues, a stable control total and a filter-propagation test. One total never proves the whole model; different checks catch different failure modes.

3. What to compare

Compare source/prepared row counts, grain, key assumptions, relationship/filter path and control totals before comparing screenshots or clicks. Equivalent valid Power BI routes are acceptable when the validation contract is satisfied.

4. Likely repair

If a control total matches but key checks fail, treat the model as untrusted. If totals differ, compare source/prepared rows and isolate the first transformation where the invariant changes.

5. Fresh retry

Close this review. Use the changed transfer task. Save a new PBIX/version/evidence row. Only a fresh successful retry can upgrade mastery.

PBM7 review - Refresh and model hygiene

1. Strong attempt should show

After a controlled source change, refresh completes and expected row/total changes are observed with stable model logic.

2. Expected reasoning

Core teacher model: Controlled source change -> Refresh -> same preparation logic replays -> validation log reconfirms invariants -> save clean PBIX/version note. Refresh is a reproducibility test. A good model can re-run its preparation steps against the same structure without manual repairs. Use clear query/table names, keep raw-source assumptions documented, avoid hidden local-file dependencies where possible, refresh after a controlled source change, and confirm row counts/control totals/relationships remain valid after refresh.

3. What to compare

Compare source/prepared row counts, grain, key assumptions, relationship/filter path and control totals before comparing screenshots or clicks. Equivalent valid Power BI routes are acceptable when the validation contract is satisfied.

4. Likely repair

If refresh breaks, identify whether the source schema/path changed, a step depends on fragile column names/types, or a relationship lost a key. Repair the reproducibility cause rather than manually fixing output rows.

5. Fresh retry

Close this review. Use the changed transfer task. Save a new PBIX/version/evidence row. Only a fresh successful retry can upgrade mastery.